

## SAFETY DATA SHEET

Revision Date 26-Mar-2024

Revision Number 3

### 1. Identification

**Product Name** Silver(II) fluoride

**Cat No. :** 11610

**CAS No** 7783-95-1  
**Synonyms** Silver(II) fluoride; Argentic Fluoride.

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### **Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                  |              |
|--------------------------------------------------|--------------|
| Oxidizing solids                                 | Category 2   |
| Acute oral toxicity                              | Category 4   |
| Acute dermal toxicity                            | Category 4   |
| Acute Inhalation Toxicity - Dusts and Mists      | Category 4   |
| Skin Corrosion/Irritation                        | Category 1 B |
| Serious Eye Damage/Eye Irritation                | Category 1   |
| Specific target organ toxicity (single exposure) | Category 3   |
| Target Organs - Respiratory system.              |              |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

May intensify fire; oxidizer  
Causes severe skin burns and eye damage  
May cause respiratory irritation  
Harmful if swallowed, in contact with skin or if inhaled

**Precautionary Statements****Prevention**

Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Wear protective gloves/protective clothing/eye protection/face protection  
Use only outdoors or in a well-ventilated area  
Do not breathe dust/fume/gas/mist/vapors/spray  
Keep away from heat/sparks/open flames/hot surfaces. - No smoking  
Keep/Store away from clothing/ other combustible materials  
Take any precaution to avoid mixing with combustibles

**Response**

Immediately call a POISON CENTER or doctor/physician

**Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

**Skin**

Wash contaminated clothing before reuse  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

**Ingestion**

Rinse mouth  
Do NOT induce vomiting

**Fire**

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

Very toxic to aquatic life with long lasting effects  
Reacts violently with water

### 3. Composition/Information on Ingredients

| Component                           | CAS No    | Weight % |
|-------------------------------------|-----------|----------|
| Silver fluoride (AgF <sub>2</sub> ) | 7783-95-1 | 99       |

### 4. First-aid measures

**General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

|                                            |                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required. Keep eye wide open while rinsing.                                                                                                                                                                          |
| <b>Skin Contact</b>                        | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Call a physician immediately.                                                                                                                                                                                                          |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Call a physician or poison control center immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. |
| <b>Ingestion</b>                           | Immediate medical attention is required. Do NOT induce vomiting. Drink plenty of water. Never give anything by mouth to an unconscious person.                                                                                                                                                                                               |
| <b>Most important symptoms and effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation                                              |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                                                                                                                                                                        |

## 5. Fire-fighting measures

|                                         |                                                                                |
|-----------------------------------------|--------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray. CO <sub>2</sub> , dry chemical, dry sand, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                       |
| <b>Flash Point</b>                      | Not applicable                                                                 |
| <b>Method -</b>                         | No information available                                                       |
| <b>Autoignition Temperature</b>         | Not applicable                                                                 |
| <b>Explosion Limits</b>                 |                                                                                |
| <b>Upper</b>                            | No data available                                                              |
| <b>Lower</b>                            | No data available                                                              |
| <b>Oxidizing Properties</b>             | Oxidizer                                                                       |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                       |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                       |

### Specific Hazards Arising from the Chemical

The product causes burns of eyes, skin and mucous membranes. Reacts violently with water. Oxidizer: Contact with combustible/organic material may cause fire. May ignite combustibles (wood, paper, oil, clothing, etc.). Do not allow run-off from fire-fighting to enter drains or water courses.

### Hazardous Combustion Products

Thermal decomposition can lead to release of irritating gases and vapors. Silver oxides. Hydrogen fluoride.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

**Health**  
3

**Flammability**  
1

**Instability**  
2

**Physical hazards**  
OX W

## 6. Accidental release measures

|                                  |                                                                                                                                                                                                                                                                         |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Use personal protective equipment as required. Evacuate personnel to safe areas. Avoid contact with skin, eyes or clothing.                                                                                                                                             |
| <b>Environmental Precautions</b> | Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. Should not be released into |

the environment.

**Methods for Containment and Clean Up** Sweep up and shovel into suitable containers for disposal. Avoid dust formation. Do not expose spill to water. Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Sweep up and shovel into suitable containers for disposal.

## 7. Handling and storage

**Handling** Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe dust. Do not ingest. If swallowed then seek immediate medical assistance. Do not allow contact with water. Handle under an inert atmosphere. Keep away from clothing and other combustible materials.

**Storage.** Corrosives area. Keep away from water or moist air. Store under an inert atmosphere. Protect from moisture. Keep containers tightly closed in a dry, cool and well-ventilated place. Do not store near combustible materials. Incompatible Materials. Organic materials. Combustible material. Reducing Agent. Bases. Strong reducing agents.

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component                           | ACGIH TLV                  | OSHA PEL                             | NIOSH                       | Mexico OEL (TWA)                                      |
|-------------------------------------|----------------------------|--------------------------------------|-----------------------------|-------------------------------------------------------|
| Silver fluoride (AgF <sub>2</sub> ) | TWA: 2.5 mg/m <sup>3</sup> | (Vacated) TWA: 2.5 mg/m <sup>3</sup> | IDLH: 250 mg/m <sup>3</sup> | TWA: 0.1 mg/m <sup>3</sup> TWA: 2.5 mg/m <sup>3</sup> |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures** Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

### Personal Protective Equipment

**Eye/face Protection** Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection** Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended Filter type:** Particulates filter conforming to EN 143.

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                     |                          |
|---------------------|--------------------------|
| Physical State      | Powder Solid             |
| Appearance          | Dark brown               |
| Odor                | No information available |
| Odor Threshold      | No information available |
| pH                  | Not applicable           |
| Melting Point/Range | 690 °C / 1274 °F         |
| Boiling Point/Range | No information available |

|                                        |                          |
|----------------------------------------|--------------------------|
| Flash Point                            | Not applicable           |
| Evaporation Rate                       | Not applicable           |
| Flammability (solid,gas)               | No information available |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | No information available |
| Vapor Density                          | Not applicable           |
| Specific Gravity                       | 4.5700                   |
| Solubility                             | Reacts with water        |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | Not applicable           |
| Decomposition Temperature              | No information available |
| Viscosity                              | Not applicable           |
| Molecular Formula                      | AgF2                     |
| Molecular Weight                       | 145.87                   |

## 10. Stability and reactivity

|                                  |                                                                                                                 |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | Yes                                                                                                             |
| Stability                        | May intensify fire; oxidizer. Oxidizer: Contact with combustible/organic material may cause fire.               |
| Conditions to Avoid              | Incompatible products. Excess heat. Combustible material. Exposure to moist air or water. Exposure to moisture. |
| Incompatible Materials           | Organic materials, Combustible material, Reducing Agent, Bases, Strong reducing agents                          |
| Hazardous Decomposition Products | Thermal decomposition can lead to release of irritating gases and vapors, Silver oxides, Hydrogen fluoride      |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                        |
| Hazardous Reactions              | None under normal processing. Reacts violently with water.                                                      |

## 11. Toxicological information

### Acute Toxicity

### Product Information

### Component Information

|                                      |                          |
|--------------------------------------|--------------------------|
| Toxicologically Synergistic Products | No information available |
|--------------------------------------|--------------------------|

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | Causes burns by all exposure routes                                                      |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component              | CAS No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|------------------------|-----------|------------|------------|------------|------------|------------|
| Silver fluoride (AgF2) | 7783-95-1 | Not listed | Not listed | Not listed | Not listed | Not listed |

|                       |                           |
|-----------------------|---------------------------|
| Mutagenic Effects     | No information available  |
| Reproductive Effects  | No information available. |
| Developmental Effects | No information available. |
| Teratogenicity        | No information available. |

|                                            |                                                                                                                                                                                                                                                            |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STOT - single exposure                     | Respiratory system                                                                                                                                                                                                                                         |
| STOT - repeated exposure                   | None known                                                                                                                                                                                                                                                 |
| Aspiration hazard                          | No information available                                                                                                                                                                                                                                   |
| Symptoms / effects, both acute and delayed | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| Endocrine Disruptor Information            | No information available                                                                                                                                                                                                                                   |
| Other Adverse Effects                      | The toxicological properties have not been fully investigated.                                                                                                                                                                                             |

## 12. Ecological information

### Ecotoxicity

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment. Reacts with water so no ecotoxicity data for the substance is available. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

|                               |                                             |
|-------------------------------|---------------------------------------------|
| Persistence and Degradability | based on information available. May persist |
| Bioaccumulation/ Accumulation | No information available.                   |
| Mobility                      | Is not likely mobile in the environment.    |

## 13. Disposal considerations

|                        |                                                                                                                                                                                                                                                                 |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Waste Disposal Methods | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 14. Transport information

### DOT

|                         |                                     |
|-------------------------|-------------------------------------|
| UN-No                   | UN3084                              |
| Proper Shipping Name    | CORROSIVE SOLIDS, OXIDIZING, N.O.S. |
| Technical Name          | Silver fluoride (AgF <sub>2</sub> ) |
| Hazard Class            | 8                                   |
| Subsidiary Hazard Class | 5.1                                 |
| Packing Group           | II                                  |

### TDG

|                         |                                    |
|-------------------------|------------------------------------|
| UN-No                   | UN3084                             |
| Proper Shipping Name    | Corrosive solid, oxidizing, n.o.s. |
| Hazard Class            | 8                                  |
| Subsidiary Hazard Class | 5.1                                |
| Packing Group           | II                                 |

### IATA

|                         |                                    |
|-------------------------|------------------------------------|
| UN-No                   | UN3084                             |
| Proper Shipping Name    | Corrosive solid, oxidizing, n.o.s. |
| Hazard Class            | 8                                  |
| Subsidiary Hazard Class | 5.1                                |
| Packing Group           | II                                 |

### IMDG/IMO

|                         |                                    |
|-------------------------|------------------------------------|
| UN-No                   | UN3084                             |
| Proper Shipping Name    | Corrosive solid, oxidizing, n.o.s. |
| Hazard Class            | 8                                  |
| Subsidiary Hazard Class | 5.1                                |
| Packing Group           | II                                 |

## 15. Regulatory information

**United States of America Inventory**

| Component              | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|------------------------|-----------|------|-----------------------------------------------|-----------------------------|
| Silver fluoride (AgF2) | 7783-95-1 | X    | ACTIVE                                        | -                           |

**Legend:**

TSCA US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

- - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b)** - Notices of Export

Not applicable

**International Inventories**

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component              | CAS No    | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|------------------------|-----------|-----|------|-----------|-------|------|------|------|-------|----------|
| Silver fluoride (AgF2) | 7783-95-1 | -   | X    | 232-037-5 | -     | X    | X    | -    | -     | KE-31275 |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**U.S. Federal Regulations****SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

| Component              | CAS No    | Weight % | SARA 313 - Threshold Values % | SARA 313 - Reporting thresholds |
|------------------------|-----------|----------|-------------------------------|---------------------------------|
| Silver fluoride (AgF2) | 7783-95-1 | 99       | 1.0 %                         | -                               |

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

| Component              | CWA - Hazardous Substances | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants |
|------------------------|----------------------------|-----------------------------|------------------------|---------------------------|
| Silver fluoride (AgF2) | -                          | -                           | X                      | -                         |

**Clean Air Act**

Not applicable

**OSHA** - Occupational Safety and Health Administration

Not applicable

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**California Proposition 65**

This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know**

## Regulations

| Component                           | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|-------------------------------------|---------------|------------|--------------|----------|--------------|
| Silver fluoride (AgF <sub>2</sub> ) | -             | X          | X            | -        | X            |

## U.S. Department of Transportation

Reportable Quantity (RQ): N  
 DOT Marine Pollutant N  
 DOT Severe Marine Pollutant N

## U.S. Department of Homeland Security

This product does not contain any DHS chemicals.

## Other International Regulations

## Mexico - Grade

No information available

## Authorisation/Restrictions according to EU REACH

Not applicable

| Component                           | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------------------------------|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Silver fluoride (AgF <sub>2</sub> ) | 7783-95-1 | -                                                                   | -                                                                             | -                                                                                                     |

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                           | CAS No    | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-------------------------------------|-----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Silver fluoride (AgF <sub>2</sub> ) | 7783-95-1 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

## Contains component(s) that meet a 'definition' of per &amp; poly fluoroalkyl substance (PFAS)?

Not applicable

## Other International Regulations

| Component                           | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-------------------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Silver fluoride (AgF <sub>2</sub> ) | 7783-95-1 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

## Prepared By

Health, Safety and Environmental Department  
 Email: chem.techinfo@thermofisher.com  
 www.thermofisher.com

## Revision Date

26-Mar-2024

## Print Date

26-Mar-2024

## Revision Summary

New emergency telephone response service provider.

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the



date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**